

Important links: [canvas](#) || [gradescope](#) || [course website](#)

Description: The significant increase in available compute over the last several decades has led to a transformation in the methods and techniques for solving science and engineering problems. To use new tools effectively, it is crucial that we understand their theoretical underpinnings and limitations, as well as practical details of their implementation. CHE 348 is a one-semester advanced undergraduate course providing students with a fundamental understanding of common numerical methods spanning chemical engineering, computational science, and applied mathematics. In problems studied, the underlying physics is emphasized throughout, with principles from *non-dimensionalization* frequently employed. While the ideas presented are general (and thus not tied to any specific programming language), the course will be carried out in [Julia](#)—a powerful, easy-to-use language designed for scientists and engineers.

Instructor: Prof. Amaresh Sahu,

- Office Hours: TBD

Teaching assistant: Albert Kyi,

- Office Hours: TBD

Attendance: Lectures are on Tuesdays and Thursdays, from 9:30–10:45 am in CPE 2.204. Discussions are on Mondays, from 1:00–3:00 pm in CPE 2.218. In-person attendance to both is mandatory; lectures and discussions will not be recorded and absences may result in a lower final grade in the course (see below).

Communication: The material and problem set questions for this course are freely available to the public at the [course website](#). All announcements will be posted to [canvas](#). Please ensure that your notification settings alert you about new announcements. In addition, students are requested to post both conceptual and problem set questions to the [canvas discussions](#) tab; do not email your content-related questions! *Both asking and answering* good questions in the [discussions](#) tab—as well as during lecture—is **highly encouraged**, and will factor into the determination of final grades (see below).

If you are emailing the instructor or teaching assistant, please include [CHE348F24] (no spaces, with square brackets) in the subject line so that the message is sorted appropriately. Emails without this text in the subject line will not be responded to.

Prerequisites: The following courses must be completed prior to starting this class, with a minimum grade of a C–:

- CHE 210—Introduction to Computing
- CHE 317—Introduction to Chemical Engineering Analysis
- M 427J—Differential Equations with Linear Algebra

The first lectures and problem set review fundamental topics from your mathematics courses, which we will build on throughout the semester.

Assignments: Assignments will be posted to the [course website](#) on Monday and due the Wednesday of the following week at 5 pm. Each assignment will consist of a coding portion, to be submitted on [canvas](#), as well as a written portion, to be submitted on [gradescope](#) (both are due at 5 pm). **Submit the coding portion as `ps##_<your-eid>.jl` where you enter the two-digit problem set number and your UT EID.** Late submissions are permitted, with a 2% penalty on the entire assignment for every hour it is late; partial hours are rounded up. Thus, for example, a submission that is on-time for the coding assignment and one second late for the written assignment will receive a 2% penalty to both—as will a submission where both coding and written portions are 59 minutes late. If you submit multiple times, then the latest time will be used to determine the penalty for the entire assignment. After 50 hours have passed since the deadline, you will receive a zero on the assignment.

Problem sets and coding assignments are an important opportunity for you to master the course material; this importance is reflected in their large weight in the final grade for the course. You are encouraged to work with your peers to solve the assigned problems. As will be the case when you enter the workplace, you are free to use resources such as [Wikipedia](#), [Mathematica](#), [WolframAlpha](#), and [StackOverflow](#). The use of large language models (LLMs) such as [ChatGPT](#) and [Perplexity AI](#) is also permitted, with an important caveat. You cannot use an LLM to generate the entire solution to a coding problem, but rather only specific sub-parts of a problem. Thus, for example, it is **never permitted** to provide an LLM with the entire coding problem statement, and then submit the generated code as your own work. At the beginning of the course, the coding tasks will be short and not contain sub-parts, in which case LLMs are not to be used. Please cite the aforementioned (and all other) resources as you use them, and remember that **you are responsible for understanding all work that you turn in**. If such resources are used and you provide an incorrect answer, partial credit will not be given. Also, electronic resources are not permitted on quizzes or exams, so a failure to understand the essential concepts will likely have negative consequences for your overall course grade.

While you are welcome to use online resources, you are **not permitted** to copy solutions from a peer, ask for answers from a student who has previously taken this class, or consult solutions from an earlier edition of this course. As already mentioned, you cannot turn in the code generated by an LLM in response to the entire problem statement as your own, and doing so constitutes cheating. Academic integrity violations will be punished severely.

Quizzes: The topics in this course build upon one another, and it is crucial that you keep up with recent material. As such, over the course of the semester there will be a total of between eight and 15 quizzes, administered at some point during lecture. These quizzes are meant to review high-level ideas, will involve minimal to no calculations, and will take roughly five minutes. Quizzes will be administered through UT InstaPoll.

Exams: There are two midterm exams and one final exam. For each midterm week, there is no problem set due. The exam schedule is as follows; missed exams will count as a zero grade (no make-up, see exceptions below). The final exam schedule is set by the registrar.

- *Midterm 1:* Mon, 30 Sep. in discussion, covering material through 19 Sep.
- *Midterm 2:* Mon, 4 Nov. in discussion, covering material through 24 Oct.
- *Final Exam:* Fri, 13 Dec., 10:30 am–12:30 pm (room TBD), covering all material

Grading: Plus (+) and minus (–) course grades will be given. At the end of the semester, raw grades will be curved to achieve an average GPA close to 3.1—a *number set by the department*. The average GPA for the class will be lower in cases of (1) collective failures of academic integrity and (2) missed exams, problem sets and/or lectures. The average class GPA will be higher if there are good discussions, both in lecture and on [canvas](#), throughout the semester. The grading breakdown is as follows:

- 30% weekly problem sets and programming assignments
- 8% in-class quizzes
- 16% midterm exam 1
- 16% midterm exam 2
- 30% final exam

Suggested reading: There is no assigned textbook, and we will rely on notes posted to the [course website](#). Additional references (i.e. websites and book chapters) will be assigned or recommended as appropriate. Since coding for the course will be done in `Julia`, you are encouraged to use any of the many excellent online references to build up your understanding of the basics. References that I recommend include:

- a gentle introduction to `Julia` [\[link\]](#)
- getting started, from the `Julia` manual [\[link\]](#)
- into to `Julia`, from the `JuliaAcademy` [\[link\]](#)
- `Julia` programming for nervous beginners, from the `JuliaAcademy` [\[link\]](#)
- From zero to `Julia`! [\[link\]](#)

Academic integrity: Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course. The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can include a written warning, probation, deferred suspension, or dismissal from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity [website](#).

Religious holy days: By UT Austin [policy](#), you must notify the course instructor of your pending absence for a religious holy day as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

Disability services: Students with disabilities may request appropriate academic accommodations from Disability and Access (D&A). If you are a student with a disability, or think you may have a disability, and need accommodations please contact D&A. You may refer to the D&A [website](#) for contact and more information. If you are already registered with D&A, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations.